IV Bolus Practice 3

Module 2 - IV Bolus Administration · Daily practice sheet · No date

Name _____ Date _____

1a 1. A 187-pound male patient received a 15 mg/kg dose of a drug by rapid IV injection. The drug is characterized by half-life of 4 hours and an apparent volume of distribution that is 20% of body weight. Determine the: a. Initial concentration of drug in the plasma

1b 1. A 187-pound male patient received a 15 mg/kg dose of a drug by rapid IV injection. The drug is characterized by half-life of 4 hours and an apparent volume of distribution that is 20% of body weight. Determine the: b. Elimination rate constant

1c 1. A 187-pound male patient received a 15 mg/kg dose of a drug by rapid IV injection. The drug is characterized by half-life of 4 hours and an apparent volume of distribution that is 20% of body weight. Determine the: c. Total body clearance

1d 1. A 187-pound male patient received a 15 mg/kg dose of a drug by rapid IV injection. The drug is characterized by half-life of 4 hours and an apparent volume of distribution that is 20% of body weight. Determine the: d. Concentration of drug in the plasma 8 hours after administration of the dose

1e 1. A 187-pound male patient received a 15 mg/kg dose of a drug by rapid IV injection. The drug is characterized by half-life of 4 hours and an apparent volume of distribution that is 20% of body weight. Determine the: e. Amount of drug in the body 12 hours after administration of the dose

2a 2. A 300-mg IV bolus dose of a drug was administered to six healthy volunteers. The parameters below best describe the pharmacokinetics of the drug.

$A = 10.21$ mg/L, $B = 8.53$ mg/L, $\alpha = 3.14$ hr^{-1}, $\beta = 0.198$ hr^{-1}

a. What is the elimination half-life in these volunteers?

2b 2. A 300-mg IV bolus dose of a drug was administered to six healthy volunteers. The parameters below best describe the pharmacokinetics of the drug.

$A = 10.21$ mg/L, $B = 8.53$ mg/L, $\alpha = 3.14\ hr^{-1}$, $\beta = 0.198\ hr^{-1}$

b. What is the initial concentration of drug in the plasma in these volunteers?

2c 2. A 300-mg IV bolus dose of a drug was administered to six healthy volunteers. The parameters below best describe the pharmacokinetics of the drug.

$A = 10.21$ mg/L, $B = 8.53$ mg/L, $\alpha = 3.14\ hr^{-1}$, $\beta = 0.198\ hr^{-1}$

c. What is the concentration of drug in the plasma 4 hours after administration of the dose?

2d 2. A 300-mg IV bolus dose of a drug was administered to six healthy volunteers. The parameters below best describe the pharmacokinetics of the drug.

$A = 10.21$ mg/L, $B = 8.53$ mg/L, $\alpha = 3.14\ hr^{-1}$, $\beta = 0.198\ hr^{-1}$

d. What is the volume of distribution of the central compartment?